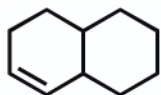


Alkene Exam Preparation Pack

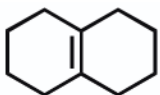
20 Essential Alkene Practice Exam Problems

Problem 1: Alkene Stability

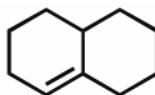
Link to answer
<http://bit.ly/Alkenes-MOC-1a>



A



B



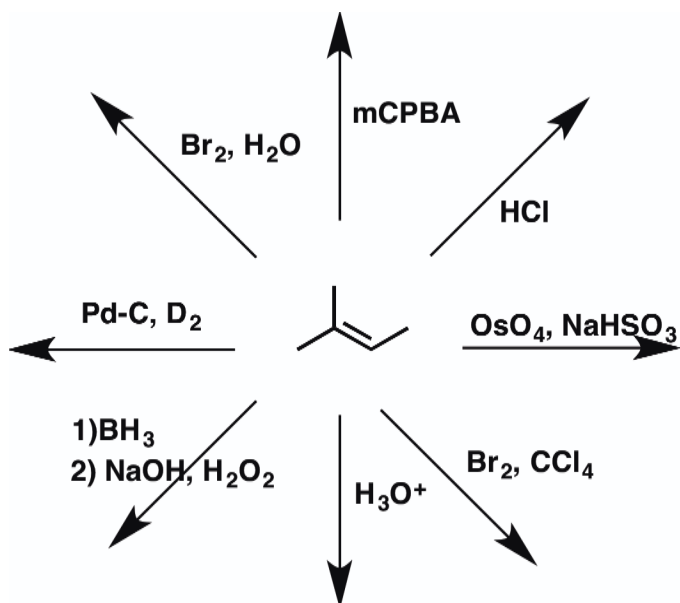
C



- a) Arrange these alkenes **A**, **B** and **C** from most stable to least stable
- b) Hydrogenation of *which alkene* would be the most exothermic?
- c) Given these three energies for ΔH° of hydrogenation (in kcal/mol):
-30.2, -28.4, -26.8
which one would be matched with -28.4 kcal/mol?

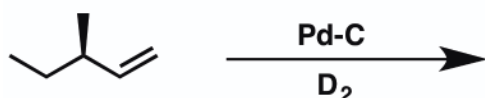
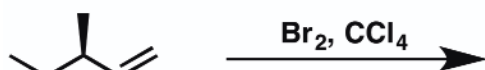
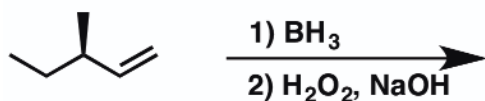
Problem 2: Draw One Product Of Each Reaction

<http://bit.ly/Alkenes-MOC-2>



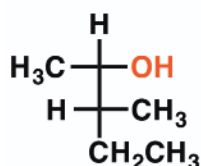
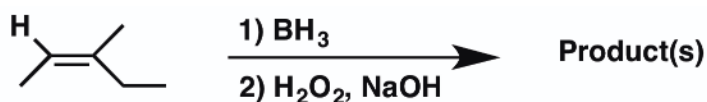
Problem 3: Provide all the products of these reactions. If there is more than one, state the relationship between the products (e.g. enantiomers, diastereomers, or constitutional isomers)

<http://bit.ly/Alkenes-MOC-3>

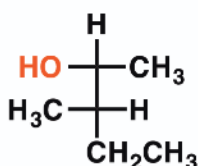


Problem 4: Pick the structure(s) which correspond to the products of the reaction below

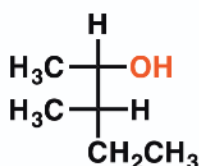
<http://bit.ly/Alkenes-MOC-4>



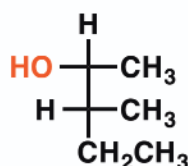
1



2

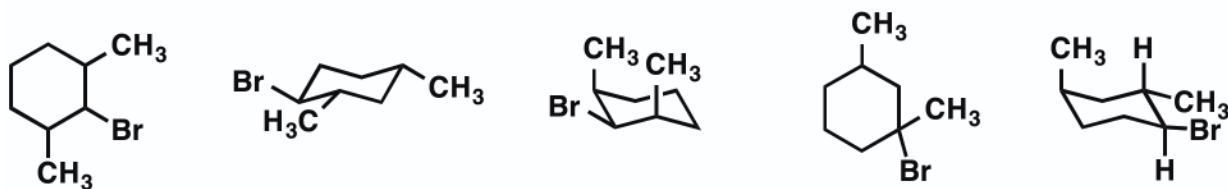
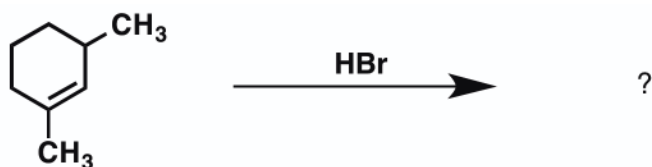


3



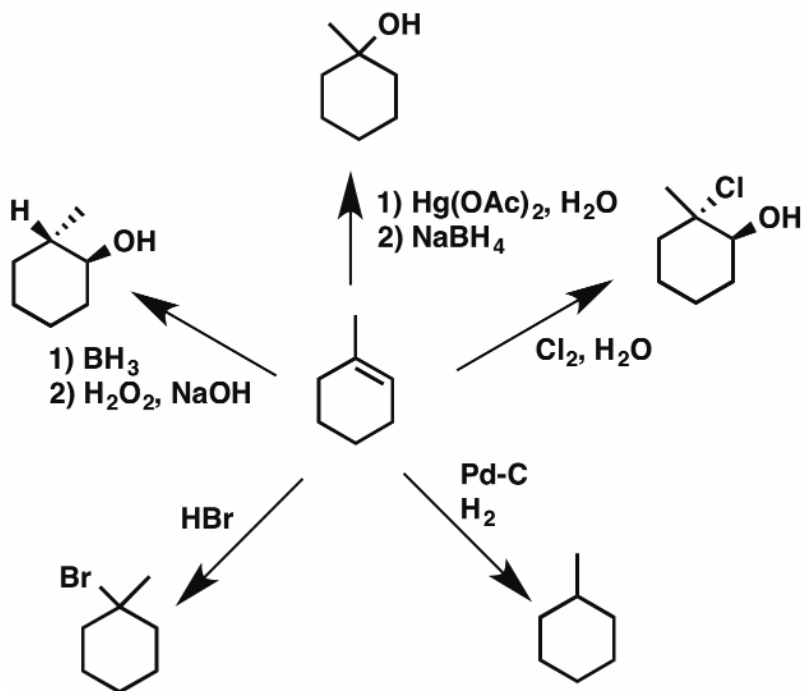
4

Problem 5: Circle the correct product of this reaction <http://bit.ly/Alkenes-MOC-5>



Problem 6: Which product is incorrect?

<http://bit.ly/Alkenes-MOC-6>



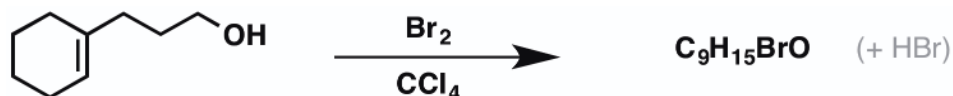
Problem 7: Draw the product(s) of the following reaction

<http://bit.ly/Alkenes-MOC-7>



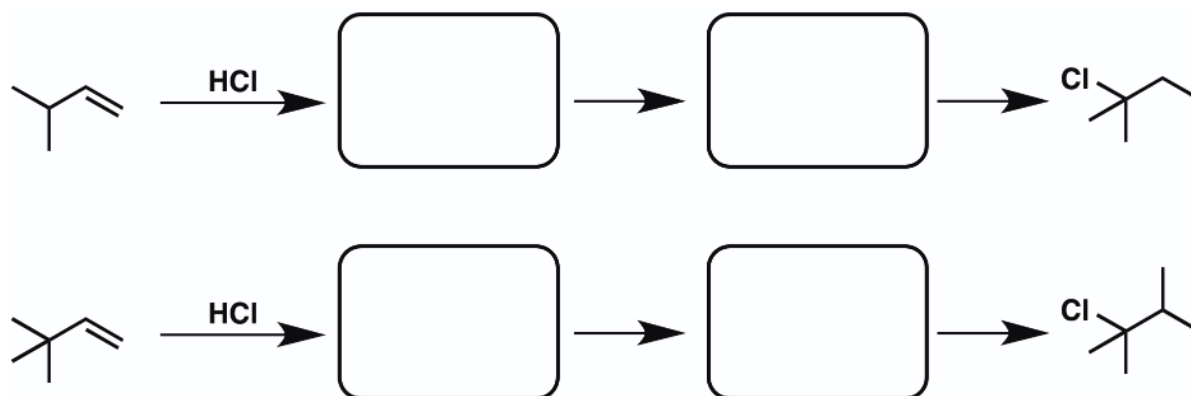
Problem 8: Draw the product of the following intramolecular reaction, and a mechanism for its formation.

<http://bit.ly/Alkenes-MOC-8>



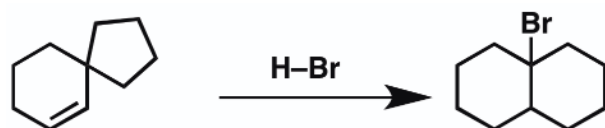
Problem 9: Draw the mechanisms for these two rearrangement reactions

<http://bit.ly/Alkenes-MOC-9>



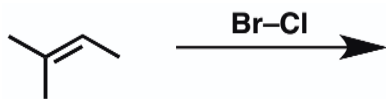
Problem 10: Draw a mechanism for this reaction

<http://bit.ly/Alkenes-MOC-10>



Problem 11: What would be the product(s) of the following reaction?

<http://bit.ly/Alkenes-MOC-11>

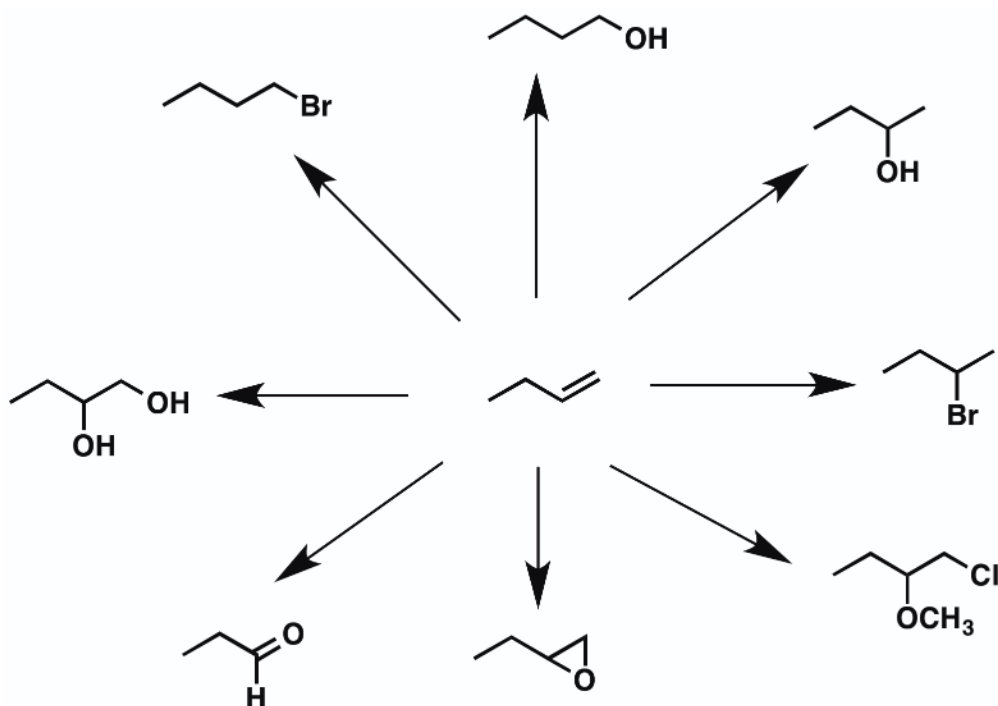


(Hint: you may not have seen this reagent before. Use electronegativity differences to figure out which atom is more electrophilic)



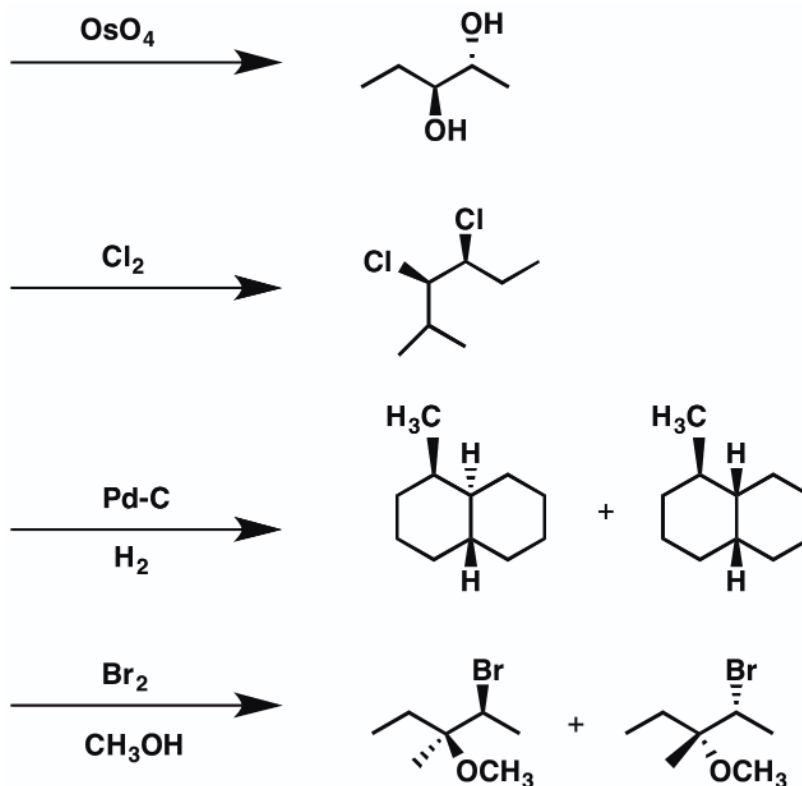
Problem 12: Give the reagents for each of the following transformations

<http://bit.ly/Alkenes-MOC-12>



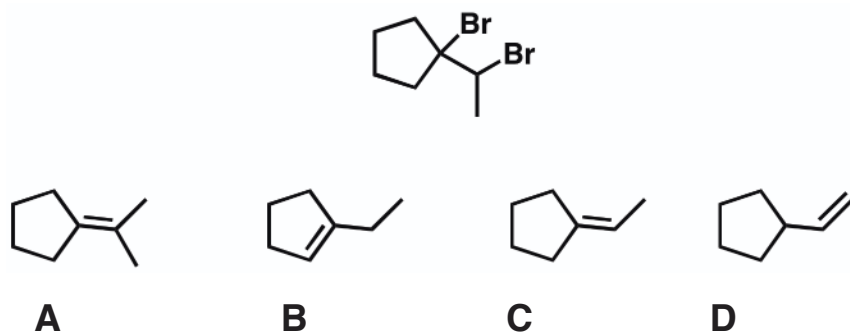
Problem 13: Draw a suitable alkene for each of the following reactions

<http://bit.ly/Alkenes-MOC-13>



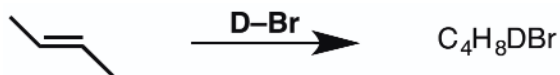
Problem 14: Which alkene (A, B, C or D) would give this dibromide product upon treatment with Br_2 and CCl_4 ?

<http://bit.ly/Alkenes-MOC-14>



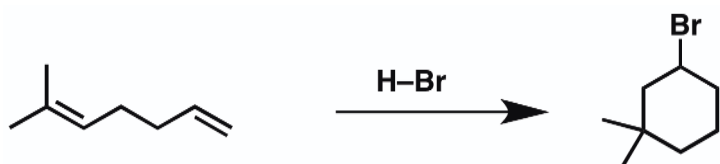
Problem 15: Draw all products resulting from this reaction and indicate how they are related to each other.

<http://bit.ly/Alkenes-MOC-15>



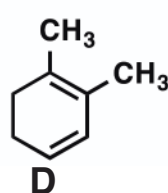
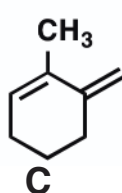
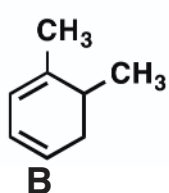
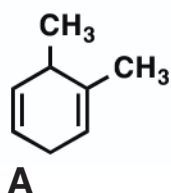
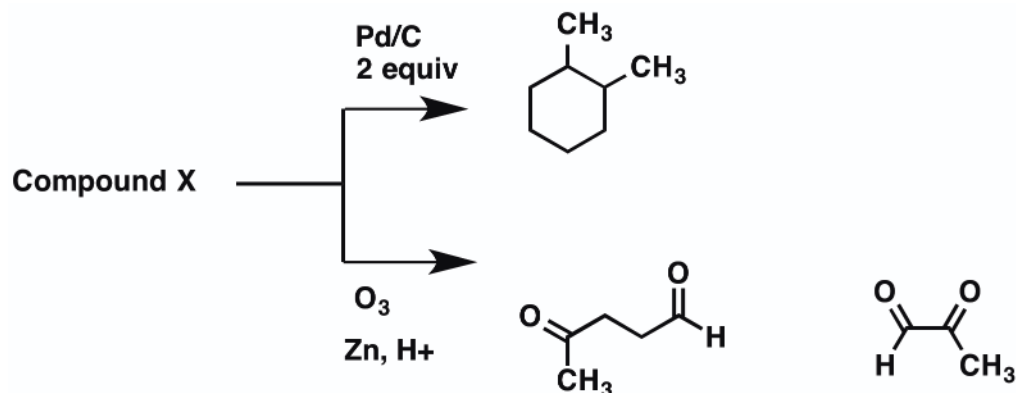
Problem 16: Draw a mechanism for this reaction:

<http://bit.ly/Alkenes-MOC-16>



Problem 17: Compound X absorbs 2 equivalents of hydrogen gas during hydrogenation with $Pd-C/H_2$. When treated with O_3 (and reductive workup) it gives the two products shown. What is compound X?

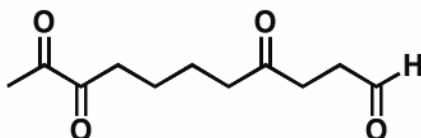
<http://bit.ly/Alkenes-MOC-17>



Problem 18: MYSTERY COMPOUND X was found to absorb 2 equivalents of H_2 gas under catalytic hydrogenation conditions. Ozonolysis of X gave the molecule shown. Draw the structure of the mystery compound

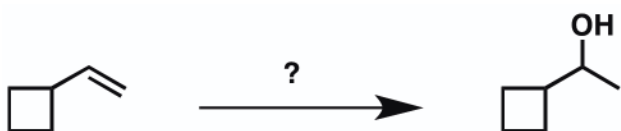


<http://bit.ly/Alkenes-MOC-18>



Problem 19: What reagent(s) would you use for the following transformation?

<http://bit.ly/Alkenes-MOC-19>



Problem 20: Draw a mechanism for the following reaction

<http://bit.ly/Alkenes-MOC-20>

